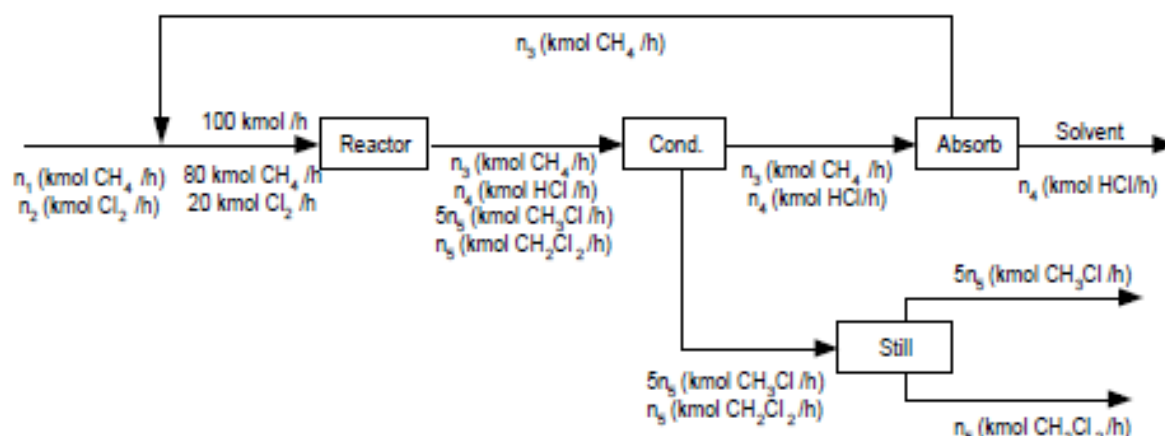


4.58 a. Basis: 100 kmol reactor feed/hr



Overall process: 4 unknowns (n_1, n_2, n_4, n_5) - 3 balances = 1 D.F.

Mixing Point: 3 unknowns (n_1, n_2, n_3) - 2 balances = 1 D.F.

Reactor: 3 unknowns (n_3, n_4, n_5) - 3 balances = 0 D.F.

Condenser: 3 unknowns (n_3, n_4, n_5) - 0 balances = 3 D.F.

Absorption column: 2 unknowns (n_3, n_4) - 0 balances = 2 D.F.

Distillation Column: 2 unknowns (n_4, n_5) - 0 balances = 2 D.F.

Atomic balances around reactor:

$$\left. \begin{array}{l} 1) \text{ C balance: } 80 = n_3 + 5n_5 + n_5 \\ 2) \text{ H balance: } 320 = 4n_3 + n_4 + 15n_5 + 2n_5 \\ 3) \text{ Cl balance: } 40 = n_4 + 5n_5 + 2n_5 \end{array} \right\} \Rightarrow \text{Solve for } n_3, n_4, n_5$$

CH₄ balance around mixing point: $n_1 = (80 - n_3)$ Solve for n_1

Cl₂ balance: $n_2 = 20$

b. For a basis of 100 kmol/h into reactor

$$\underline{n_1 = 17.1 \text{ kmol CH}_4/\text{h}}$$

$$\underline{n_2 = 20.0 \text{ kmol Cl}_2/\text{h}}$$

$$\underline{n_3 = 62.9 \text{ kmol CH}_4/\text{h}}$$

$$\underline{n_4 = 20.0 \text{ kmol HCl/h}}$$

$$\underline{5n_5 = 14.5 \text{ kmol CH}_3\text{Cl/h}}$$

c. $(1000 \text{ kg CH}_3\text{Cl/h})(1 \text{ kmol}/50.49 \text{ kg}) = 19.81 \text{ kmol CH}_3\text{Cl/h}$

$$\underline{\text{Scale factor}} = \frac{19.81 \text{ kmol CH}_3\text{Cl/h}}{14.5 \text{ kmol CH}_3\text{Cl/h}} = 1.366$$

$$\underline{\text{Fresh feed:}} \quad \left. \begin{array}{l} n_1 = (17.1)(1.366) = 23.3 \text{ kmol CH}_4/\text{h} \\ n_2 = (20.0)(1.366) = 27.3 \text{ kmol Cl}_2/\text{h} \end{array} \right\} \Rightarrow \underline{\underline{n_{\text{tot}} = 50.6 \text{ kmol/h}}}$$

$$\underline{\text{Recycle:}} \quad n_3 = (62.9)(1.366) = \underline{\underline{85.9 \text{ kmol CH}_4 \text{ recycled/h}}}$$

$$\underline{\underline{46.0 \text{ mol}\% \text{ CH}_4, 54.0 \text{ mole}\% \text{ Cl}_2}}$$